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Cryptocurrency Momentum, Reversal, Perpetuals, Open Interest, and Liquidation Cascades

Executive summary

The strongest high-confidence result in the recent literature is that crypto return continuation and reversal are highly conditional on size, liquidity, horizon, and whether one is measuring spot return dynamics or derivatives crowding. The cleanest split is this: very short-horizon reversal tends to live in smaller, thinner, lower-volume coins, while short-horizon continuation is stronger in larger, more liquid coins and in momentum signals tied to smoother information diffusion rather than one-off vertical spikes. That is why the literature often looks contradictory: equal-weighted, low-filter, small-coin samples find reversal; filtered, value-weighted, larger-coin samples find momentum. [1]

For a small trading firm looking for overlooked alpha in the user-defined “small-cap” bucket, a bottom-20%-by-market-cap implementation is not directly tested in most papers, but the closest evidence points to three families worth taking seriously: short-term liquidity-provision reversal in low-volume names, cross-coin information-diffusion signals at very short intraday horizons, and post-deleveraging reversal after genuine liquidation flushes. By contrast, pure breakout continuation in small caps is much less robust than discretionary crypto lore suggests; the better evidence is for continuation in large/liquid names or in “smooth-path” moves, while parabolic one-shot spikes in small caps more often mean-revert. [2]

Funding rates and open interest do contain information, but mostly as conditional state variables, not as stand-alone “buy when X / sell when Y” predictors. Academic and institutional work supports three narrower claims: crypto carry or basis/funding premia exist and are linked to segmentation and demand imbalances; funding and open interest have predictive content for Bitcoin and some perp markets; and raw funding/liquidation series can be too slow or incomplete unless combined with price and open-interest changes. In practice, price-plus-open-interest is more useful than open interest alone, and funding is more useful as a crowding and carry-cost variable than as a universal directional signal. [3]

Implementation frictions are not a side note; they change the ranking of these alphas. Realistic studies that include fees, slippage, daily mark-to-market, and liquidation risk show that many “significant” momentum portfolios either lose economic significance or become operationally dangerous, especially on the short side. This matters even more in small caps, where exchange fragmentation, thin books, wider spreads, per-user open-interest caps, KYC frictions, and venue-specific liquidation engines can easily overpower paper alpha. [4]

My bottom-line ranking is: for small-cap crypto, the most plausible overlooked alpha is short-horizon reversal after liquidity shocks and liquidation-driven overshoots; the most scalable alpha is large/liquid breakout or high-momentum continuation; funding and open interest belong in regime filters and risk overlays; and “parabolic reversal” is real enough to respect, but the literature supports it mainly as an overreaction/liquidity event, not as a simple chart pattern that works unchanged across the coin spectrum. [5]

What the evidence says by signal family

Momentum and breakout continuation

At the broad asset-class level, momentum is established in crypto, but not in a uniform way. Liu, Tsyvinski, and Wu show that market, size, and momentum form a three-factor structure for cryptocurrency returns, and their momentum factor is built from three-week past performance sorted weekly. That is evidence that continuation is not just anecdotal; it exists strongly enough to show up as a priced common factor. [6]

The more recent "realistic assumptions" work is more cautious. Han, Kang, and Ryu use a survivorship-bias-free CoinMarketCap universe from December 28, 2013 through August 28, 2023, apply size and volume filters, estimate actual fees, tick-size-driven transaction costs, and slippage from trading data, and then mark positions to market daily. Their conclusion is that time-series momentum is much stronger than cross-sectional momentum, that the effect is concentrated in large winners, and that losers often rebound violently enough to wreck short legs. They explicitly argue that the simple t-test of mean return is not enough in crypto because jump risk and liquidation risk can make "significant" strategies operationally poor. [7]

This bears directly on Qullamaggie-style breakouts. The best direct analogue in the literature is not "breakout above base on volume" language, but "distance from prior high" and path-dependent continuation. Fičura's weekly study splits coins into "large and liquid" versus "small and illiquid," using 50 million USD market cap and 5 million USD weekly volume thresholds over 2017/06–2022/12, and finds that "high momentum," measured as distance from the prior k-week high, is a stronger predictor than plain momentum. On large/liquid coins, distance from the one-week high predicts future returns positively with a Newey-West t-stat of 4.93; on small/illiquid coins, the same signal predicts negatively with a t-stat of -9.03. In other words, breakout continuation is supported for liquid leaders and contradicted for smaller, thinner names. [8]

A related but still preliminary result comes from Woong Bae Kim's 2026 SSRN paper on price-path continuity. Using a broad survivorship-mitigated sample from January 2020 to April 2026, the paper finds that short-term past returns reverse on average, but the same past returns become more likely to continue when they were built through smoother, more continuous price paths rather than discrete jumps. That is the closest academic support for "episodic pivot" or "clean trend ignition" logic: steady accumulation appears more predictive than one-bar excitement. It is not yet peer-reviewed, but the mechanism is coherent and consistent with limited-attention theory. [9]

Intraday continuation also exists. Wen, Bouri, Xu, and Zhao examine Bitcoin from March 3, 2013 to May 31, 2020 using hourly data and find both intraday momentum and reversal; the pattern varies with jumps, FOMC releases, liquidity, and COVID. Their evidence supports very short-run continuation under some states, but not a simple always-on breakout rule. [10]

The best reading, then, is mixed but usable. Continuation breakouts have support in large/liquid crypto and in smoother trend formations; they are much weaker, and often inverted, in small/illiquid names. For the bottom-20%-by-cap bucket, the evidence does not support porting a large-cap breakout playbook unchanged. [11]

Short-term reversal and liquidity-provision alpha

The clearest small-cap result in the literature is short-term reversal. Bianchi, Babiak, and Dickerson's Riksbank working paper studies daily data from March 1, 2017 to March 1, 2022 and dynamically selects the top 100 most liquid crypto pairs. They show a significant short-term reversal, then strengthen it by conditioning on de-trended volume: expected returns from this liquidity-provision proxy are concentrated in low-volume pairs, and the equal-weighted low-volume conditional reversal strategy remains positive and significant after their conservative transaction-cost assumptions, with average daily return reported at 1.68% and p-value 0.001. They also show that top trading activity is extremely concentrated: the top 10% of assets account for roughly 90% of trading volume, implying the long tail is thin enough that liquidity-provision premia can survive there longer. [12]

That result lines up almost perfectly with the "small-cap overlooked" angle. In low-volume crypto, the alpha looks less like classic momentum and more like being paid to absorb temporary order-flow imbalances. The caveat is just as important: Bianchi's transaction-cost treatment is simplified, and the strategy is an end-of-day proxy for liquidity provision, not a literal real-time market-making recipe. [13]

Fičura sharpens the size split at weekly frequency. Using CoinMarketCap data over 2017/06–2022/12, he finds that the previously documented one-week reversal is mostly a small/illiquid phenomenon with t-stat -7.31, while large/liquid coins show one-week momentum with t-stat 2.33. He further notes that the small-cap reversal is driven mostly by very low-volume coins, often below $100,000 in last-week volume, which are hard to buy or sell at observed closing prices. This is exactly the kind of pattern larger firms tend to ignore because the trade is not wrong but too capacity-constrained, too operationally messy, or both. [14]

Other studies help explain why the literature disagrees. Fičura's review notes that older datasets, equal-weighted portfolios, and no minimum market-cap filters tend to find reversal, while newer datasets with market-cap filters and value-weighting tend to find momentum. That is less a contradiction than a sample-selection map. If you want bottom-decile or bottom-quintile-by-cap alpha, you should expect reversal to dominate. If you filter heavily for tradability, continuation reappears. [8]

There is also evidence of extreme-event overreaction. Caporale and Plastun, using daily data on Bitcoin, Litecoin, Ripple, and Dash from April 28, 2013 to December 31, 2017, find that next-day price changes after overreaction days are larger than after normal days. But their trading simulations do not produce statistically convincing exploitability for the contrarian rule once the event is translated into trades. That is useful because it distinguishes "there is a pattern" from "there is harvestable alpha." [15]

The practical reading is that short-term reversal is probably the strongest overlooked signal in small-cap crypto, but it is a microstructure trade, not a clean factor. It depends on access to the right venue, patient execution, and a willingness to skip names where apparent alpha is just stale closing marks painted onto a shallow book. [16]

Funding rates, basis, and open-interest signals

Funding rates matter, but mostly as a measure of crowding, carry cost, and leverage imbalance. The BIS "Crypto carry" paper by Schmeling, Schrimpf, and Todorov documents persistent positive basis and carry in BTC and ETH futures, links it to institutional segmentation and the difficulty of arbitrage capital moving between venues, and shows that nonreportable and institutional positioning on CME correlates with basis changes. Their daily sample covers March 2019 to July 2024 for the main futures-basis series. The average annualized one-month basis in their table is economically large: 8.21% on OKEx BTC one-month, 6.41% on CME BTC one-month, and similarly high on ETH. That is strong evidence that derivatives demand imbalances persist long enough to matter. [17]

But carry is not a free lunch. The same BIS paper shows that leverage makes the trade path-dependent and fragile: in their examples, with leverage of 5 the futures leg goes bankrupt 31% of the time before maturity, at leverage 10 it is 52%, and at leverage 20 it is 71%. This is a clean reminder that funding and basis premia often coexist with large interim mark-to-market losses. [17]

Academic evidence on predictive power is narrower than crypto Twitter would suggest. Palazzi, Júnior, and Klotzle's 2026 SSRN paper argues that crypto-native valuation and positioning variables, especially MVRV, funding rates, and open interest, predict Bitcoin returns in every regime from 2014 to 2024, while macro-finance variables matter more after 2019 and especially in 2022–2023. That supports funding and open interest as real predictive inputs, though the result is for Bitcoin and at regime level, not for the small-cap cross-section. [18]

The more practical problem is measurement. Glassnode argues that funding rates can react too slowly during rapid deleveraging and can remain subdued even during large position unwinds if spot and perp prices move together. They recommend combining price and open interest, not relying on funding or liquidation data alone. Their "price/OI matrix" is intuitive: price up plus OI up suggests new longs; price down plus OI up suggests new shorts; price down plus OI down suggests longs closing; price up plus OI down suggests shorts closing. [19]

That approach is strengthened by data-quality research. Giagkiozis and Said's 2024 Ledger article shows that reported Bitcoin perp open interest is systematically misquoted by some major exchanges, and that some venues appear to delay liquidation-related messages. They identify these issues using tick-by-tick data across seven large exchanges during two 2023 periods. For anyone building open-interest signals in smaller-cap perps, this is not an academic footnote; it is a direct warning that vendor-normalized OI series may embed venue-specific errors. [20]

My synthesis is that funding and open interest do have predictive value, but their best use is conditional. Positive funding with rising open interest into a falling spot market is often a crowded-long warning. Negative funding after a washout may point to a squeezeable short base. Rising OI with falling price is new shorting, not necessarily capitulation. And on small-cap alts, coverage is sparse enough that raw funding/OI signals are weaker than in BTC/ETH/SOL. [21]

Liquidation cascades and reversal after forced deleveraging

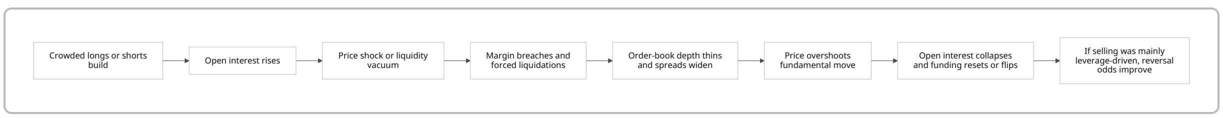
Liquidation-cascade reversals are real market events, but the academic literature is still thin and mostly case-study based. The strongest current evidence comes from practitioner and event-study material. FalconX's January 31, 2026 note describes a selloff in which BTC first triggered long liquidations, then saw open interest rise as traders added shorts, then went through further liquidation waves under $80,000 before bottoming near $75,000 and stabilizing around $78,000. Funding moved from positive to negative by the end of the day, while ETH and SOL funding flipped deeply negative during the flush. That sequence is exactly what a tradable liquidation-reversal setup looks like in practice: initial forced unwind, crowd inversion, then overshoot. [22]

Galaxy’s February 2025 market commentary describes a related pattern in ETH: more than $1 billion in long liquidations, ETH open interest dropping from nearly $5 billion to under $4 billion in a few days, and a general “reset of market leverage.” K33’s August 2023 note makes the same point more explicitly: after one of the largest relative daily open-interest declines since March 2021, market exposure to near-term squeezes had been materially reduced. In both cases, the alpha is not “buy because price fell,” but “buy when leverage has been forcibly removed and marginal sellers are exhausted.” [23]

Glassnode adds an important distinction: not every sharp drop is a one-sided long liquidation. When spot and futures cumulative volume delta fall together and open interest compresses, the move may reflect both real-money selling and leveraged deleveraging, not just a reflexive cascade. That matters because liquidation-cascade reversal works best when the move is primarily leverage-driven. If the drop is spot-led fundamental supply, negative funding alone is not enough. [24]

A recent SSRN wave is beginning to formalize this topic. Recent abstracts on minute-level crypto cascades and hidden-liquidity execution risk study the October 2025 crash using exchange-level open-interest and deep order-book data, emphasizing reflexive leverage/liquidity loops and the mismatch between displayed and true executable depth. Those papers are too new to treat as settled, but they fit the practitioner record and the microstructure logic. [25]

The conclusion here is stronger than for funding and weaker than for plain reversal. Liquidation-cascade reversals are a real edge, but only as an event-driven setup that demands confirmation from price, open interest, and funding together. They are not a mechanical daily factor. [26]



The flow above is a synthesis of the price/OI interpretation framework from Glassnode and the event narratives from FalconX, Galaxy, and K33. The key practical point is that the reversal signal strengthens only after leverage is demonstrably removed from the system. [27]

Concise synthesis of key papers and reports

Source	Sample and asset selection	Frequency and method	One-line takeaway
Liu, Tsyvinski, Wu, *Journal of Finance* 2022	Broad cross-section of cryptocurrencies; factor sorts by size and three-week momentum	Weekly factor construction	Momentum is strong enough to appear as a common priced factor alongside market and size in crypto returns. [6]

Source	Sample and asset selection	Frequency and method	One-line takeaway
Han, Kang, Ryu, SSRN 2024	All CoinMarketCap coins, survivorship-bias-free; 2013-12-28 to 2023-08-28; liquid/futures subsets added	Daily, with fees, tick sizes, slippage, mark-to-market, liquidation accounting	Time-series momentum is strong, cross-sectional momentum is weak, and many “significant” strategies break once realistic frictions and liquidation risk are included. [28]
Bianchi, Babiak, Dickerson, Riksbank WP 2022	Top 100 most liquid pairs; 2017-03-01 to 2022-03-01	Daily reversal and double-sort on de-trended volume	Short-term reversal is strongest in low-volume pairs, consistent with liquidity-provision alpha rather than broad factor exposure. [29]
Guo, Sang, Tu, Wang, *JEDC* 2024	Binance cross-section	Minute-level predictive models, adaptive LASSO, PCA, OOS tests	Cross-coin lagged returns predict focal-coin returns up to about ten minutes, and the long-short strategy remains positive out of sample after costs. [30]
Wen, Bouri, Xu, Zhao, *NAJEF* 2022	Bitcoin 2013-03-03 to 2020-05-31; also ETH, LTC, XRP	Hourly intraday predictability	Crypto shows both intraday momentum and reversal, with the mix shifting across jumps, FOMC days, liquidity states, and COVID. [10]
Borgards, *NAJEF* 2021	21 Bitfinex assets, 2014-01-01 to 2019-12-31	Daily, hourly, 5-minute; turning-point dynamics	Time-series momentum exists as a dynamic sequence of trend phases; smoother persistence matters more than arbitrary lookbacks. [31]
Fičura, Prague WP 2023	CoinMarketCap universe; 2017/06–2022/12; split into large/liquid vs small/illiquid	Weekly momentum, reversal, and distance-from-high signals	Weekly reversal belongs to small/illiquid coins, while large/liquid coins show momentum; distance from prior high is a superior continuation signal in liquid names and an anti-signal in small/illiquid names. [14]
Proelss, Schweizer, Buchwalter, SSRN 2024 / FRL 2025	Survivorship-bias-free sample, attention to market-cap differences	Portfolio tests by size segment	Effective momentum strategies appear among larger-cap cryptocurrencies, not where many retail traders expect them. [32]

Source	Sample and asset selection	Frequency and method	One-line takeaway
Schmeling, Schrimpf, Todorov, BIS / *Management Science*	BTC and ETH futures across major exchanges; 2019-03 to 2024-07	Daily basis/carry, positions, and leverage-path analysis	Crypto carry exists and is economically large, but leverage makes it fragile and path-dependent. [33]
Brauneis, Mestel, Riordan et al., *Journal of Empirical Finance* 2022	BTCUSD on Bitfinex, Bitstamp, Coinbase Pro; 2017-12-15 to 2019-12-15	Order-book microstructure, spreads, price impact, roundtrip cost	Large-cap BTC can be remarkably cheap to trade on major venues, but liquidity and depth still vary by venue and by time of day. [34]
Giagkiozis, Said, *Ledger* 2024	Tick-by-tick BTC perpetuals across seven major exchanges in 2023	OI/volume reconciliation	Some exchanges misquote or delay open-interest and liquidation information, which directly weakens naive open-interest alpha. [20]
Palazzi, Júnior, Klotzle, SSRN 2026	Bitcoin, five regimes from 2014 to 2024	Frequency-domain Granger causality and transfer entropy	Funding rates and open interest predict Bitcoin returns across regimes, but this is a BTC-regime result, not yet a small-cap playbook. [18]
Caporale, Plastun, CESifo 2018	BTC, LTC, XRP, Dash; 2013-04-28 to 2017-12-31	Daily overreaction-event tests and trading simulation	Extreme-day overreaction patterns exist, but turning them into robust profit is harder than the anomaly tables imply. [35]

Comparative table of signals

The table below is a synthesis, not a direct copy of any single source. “Expected alpha” means evidence-weighted plausibility after implementation frictions, not gross backtest return. The small-cap angle refers to the user-specified convention of treating the bottom 20% by market cap as small-cap when a paper does not define it exactly.

Signal family	Where the evidence is strongest	Expected alpha after realistic frictions	Turnover	Required capital	Typical transaction-cost burden	Tail-risk / drawdown profile	Evidence basis
Short-horizon liquidity-provision reversal	Small/ illiquid, low-volume coins; also post-event overreaction days	**High for sub-scale firms**, but capacity-limited	High	Low to medium	High in small caps; cost assumptions often dominate	Moderate to high; execution risk more than factor crash	Bianchi low-volume reversal; Fičura weekly small-cap reversal; Caporale-Plastun event patterns. 36
Large/liquid breakout continuation	BTC/ETH/ top-liquidity alts, filtered universes	**Moderate to high**, more scalable	Medium	Medium to high	Moderate on major venues	High crash risk on short leg and during momentum unwind	Han realistic-assumption momentum; Fičura high-momentum-by-week-high; Liu-Tsyvinski-Wu factor evidence. 37
Small-cap breakout continuation	Bottom tail / thin books / jumpy coins	**Low to mixed**	Medium to high	Low	High	High gap and reversal risk	Fičura finds breakout-style signal flips sign in small/illiquid names; smooth-path continuation helps more than discrete spikes. 38
Cross-coin lead/lag continuation	Exchange-level intraday, especially Binance	**Moderate**, operationally demanding	Very high	Low to medium	Moderate if automated and internalized well	Lower crash risk than momentum, higher tech/ execution risk	Guo et al. find up-to-10-minute predictability and positive OOS returns after costs. 30

Signal family	Where the evidence is strongest	Expected alpha after realistic frictions	Turnover	Required capital	Typical transaction-cost burden	Tail-risk / drawdown profile	Evidence basis
Funding-rate directional signal alone	Mostly BTC/ETH/SOL perps	**Low to moderate**	Low	Medium	Low trading cost, but carry cost matters	Can stay crowded for long periods	Funding predicts in some regime studies, but Glassnode shows it can lag deleveraging. [39]
Funding-plus-price crowding filter	Major perps; crowded-long or crowded-short states	**Moderate**, best as filter	Low to medium	Medium	Low to moderate	Good at avoiding bad states rather than printing standalone alpha	Glassnode, FalconX, Galaxy, Block Scholes practitioner framework. [40]
Open-interest change alone	Major perps only	**Low**	Low	Medium	Low	Ambiguous without price context	Raw OI is directionally ambiguous and data quality can be poor. [41]
Price-plus-open-interest matrix	Major perps and event windows	**Moderate to high** as positioning-state filter	Medium	Medium	Low to moderate	Strong for event diagnosis; weaker as a daily standalone factor	Glassnode LPOC framework; FalconX January 2026 cascade. [42]
Liquidation-cascade reversal	Genuine leveraged flushes with OI collapse and funding reset	**High when selective**, very poor if used mechanically	Event-driven	Low to medium	Moderate; spread widens when signal is strongest	Very high if you front-run the flush instead of waiting for confirmation	FalconX, Galaxy, K33, recent minute-level SSRN case studies. [43]

Signal family	Where the evidence is strongest	Expected alpha after realistic frictions	Turnover	Required capital	Typical transaction-cost burden	Tail-risk / drawdown profile	Evidence basis
Cash-and-carry / basis harvesting	BTC/ETH and major-exchange futures	**Moderate**, but risk-adjusted not trivial	Low	Medium to high	Low trading cost, but margin financing and path losses matter	Very high under leverage; funding/carry can be swamped by path risk	BIS crypto carry. [17]

Practical implementation notes for small trading firms

A small firm should separate two businesses that are often mixed together in crypto discourse: scalable continuation trading in large liquid names, and inventory-absorbing reversal trading in thin names. The first behaves like a standard systematic momentum sleeve with all the usual crash problems. The second behaves more like market making or event-driven liquidity provision, where the edge sits in execution quality, quote patience, and selective participation rather than in simple signal strength. The literature is clearest on this split, even if it uses different terminology. [44]

For small-cap spot alphas, a firm should assume that reported close-to-close returns are optimistic. Brauneis et al. show that even in BTC on major exchanges, liquidity is venue- and depth-dependent; their low roundtrip estimates for a $100,000 BTC trade on Coinbase Pro and Bitfinex do **not** transfer to the small-cap tail. Fičura explicitly warns that much of the small/illiquid reversal effect is driven by coins with very low trading volume, where getting the recorded closing price is doubtful. That means any serious backtest should mark to next-bar mid, haircut fills by queue position, and cap participation by a small percentage of visible and inferred depth. [45]

Perpetual-futures implementation has its own frictions. On Binance, the default funding interval is every eight hours, though funding intervals can be adjusted in stressed markets. Larger notionals face lower allowable leverage via risk tiers, and leverage adjustment requires KYC. On Bybit, liquidation is triggered off mark price rather than last price, risk-limit tiers tighten position size as leverage rises, and per-user open positions can be capped as a percentage of contract open interest. Those rules matter because a strategy that looks neutral at signal level can still fail operationally if KYC, risk tiers, or OI caps bind at the wrong time. [46]

For funding and OI signals, the best practice is to build a state machine rather than a directional model. A practical version is: rising price and rising OI means new longs, so avoid fading too early; falling price and rising OI means fresh shorting, so do not call “capitulation”; falling price and falling OI is closer to long liquidation; rising price and falling OI is short covering. Funding then acts as a cost-of-crowding overlay: very positive funding plus rising OI into weakening spot is a warning, while very negative funding after a sharp OI collapse can improve reversal odds. This uses the strengths of each metric without depending on any single noisy series. [26]

If the firm wants one concrete research agenda for the small-cap universe, it should prioritize four tests. First, a maker-style one- to three-day reversal model on bottom-quintile-by-cap coins with strict participation caps and venue exclusions. Second, a cascade-reversal event study requiring simultaneous price drop, open-interest drop, and funding reset rather than price alone. Third, a cross-coin lead/lag model restricted to a single venue to reduce synchronization errors. Fourth, a breakout-continuation model that excludes "parabolic" one-bar spikes and instead ranks on smooth-path continuity or distance from recent highs only when depth and spread pass filters. Those are the places where the literature is most supportive and where larger firms are least likely to deploy meaningful balance sheet. [47]

Recommended further reading

The list below is ordered by source credibility and direct usefulness for the questions you asked.

Highest-priority academic sources

Source	Why it matters
Liu, Tsyvinski, Wu, *Common Risk Factors in Cryptocurrency*, *Journal of Finance* 2022	Best anchor for crypto momentum as a priced factor rather than a chart trope. [48]
Han, Kang, Ryu, *Time-Series and Cross-Sectional Momentum in the Cryptocurrency Market: A Comprehensive Analysis under Realistic Assumptions*, SSRN 2024	Best paper on what survives once you include tradability, slippage, fees, margin, and liquidation accounting. [49]
Guo, Sang, Tu, Wang, *Cross-cryptocurrency return predictability*, *Journal of Economic Dynamics and Control* 2024	Best evidence for short-horizon cross-coin information-diffusion alpha. [50]
Wen, Bouri, Xu, Zhao, *Intraday return predictability in the cryptocurrency markets*, *North American Journal of Economics and Finance* 2022	Best single paper on when intraday momentum and reversal coexist. [51]
Brauneis, Mestel, Riordan et al., *Bitcoin unchained: Determinants of cryptocurrency exchange liquidity*, *Journal of Empirical Finance* 2022	Best microstructure paper here for spreads, depth, price impact, and time-of-day liquidity. [52]
Li, Urquhart, Wang, Zhang, *MAX momentum in cryptocurrency markets*, *International Review of Financial Analysis* 2021	Important for crash-prone "lottery" or parabolic behavior in crypto. [53]

Strong working papers and SSRN/arXiv sources

Source	Why it matters
Schmeling, Schrimpf, Todorov, *Crypto carry*, BIS WP / later *Management Science*	Best source on basis/funding premia and their structural drivers. [33]

Source	Why it matters
Proelss, Schweizer, Buchwalter, *Do Risk Preferences Drive Momentum in Cryptocurrencies?*	Useful corrective against the idea that momentum is strongest where risk is highest; their result points toward larger caps. [32]
Palazzi, Júnior, Klotzle, *The Evolving Predictability of Bitcoin Returns*	Useful for funding and OI as regime-varying Bitcoin predictors. [18]
Woong Bae Kim, *Price Path Continuity and the Cross-Section of Cryptocurrency Returns*	Most relevant recent source for episodic pivots and smoother-path continuation. Not peer-reviewed yet. [9]
Giagkiozis, Said, *Reconciling Open Interest with Traded Volume in Perpetual Swaps*, *Ledger* 2024	If you use OI data operationally, read this before trusting vendor feeds. [20]

Practitioner and institutional research worth reading

Source	Why it matters
Glassnode, *Inferring Leveraged Positioning from Price and Open Interest*	Best practical framework for interpreting OI with price, and best critique of raw funding/liquidation data. [19]
FalconX, *Unpacking the January 31 Liquidation Cascade*	Clear event anatomy of a real cascade and reversal sequence. [22]
Galaxy, *Price Action Triggers Liquidation Cascade, Slashing Funding Rates and Wiping Out Open Interest*	Good for leverage-reset mechanics and how OI compression changes market state. [54]
K33 research archive on OI, funding, and leverage shakeouts	Useful for discretionary timing language around crowding and leverage resets. [55]
Binance Research and official derivatives docs	Useful for implementation constraints, funding intervals, and venue-specific mechanics. [56]

Open questions and limitations

The biggest limitation in the literature is that “small-cap crypto” is rarely analyzed as an exact bottom-20%-by-market-cap slice with realistic execution. Most papers either use all coins, equal-weighted cross-sections, minimum-cap filters, or ad hoc “large/liquid vs small/illiquid” partitions. So the small-cap conclusions above are best read as evidence-weighted proxies, not as exact estimates for the bottom quintile. [57]

The second limitation is data quality in derivatives. Funding formulas can change across venues and through time, open-interest reporting is not fully reliable, liquidation data are incomplete, and true executable depth during stress can diverge sharply from displayed depth. That is precisely why recent practitioner work puts more weight on multi-metric state diagnosis than on single-series signals. [58]

The third limitation is that direct academic evidence for discretionary chart concepts such as “Qullamaggie-style continuation breakout,” “episodic pivot,” and “parabolic reversal” is mostly indirect. The literature supports some underlying mechanisms—distance from highs, smooth-path continuity, overreaction, limited attention, and forced deleveraging—but it does not validate the full discretionary package as one coherent standardized alpha. In crypto, the pattern may be visible on the chart while the edge actually lives in something more mechanical: path smoothness, crowding state, or the absence of liquidity. [59]

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